

# CM0859 – MT5009 Type Theory

## Appendix: Testing with QuickCheck

Andrés Sicard-Ramírez

Universidad EAFIT

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# Testing with QuickCheck

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## A paper

Koen Claessen and John Hughes (2000). QuickCheck: A Lightweight Tool for Random Testing of Haskell Programs. ICFP'00. DOI: <https://doi.org/10.1145/357766.351266>.

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## A paper

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## Most Influential ICFP Paper Award 2010<sup>†</sup>

*“The techniques described in the paper have spawned a significant body of follow-on work in test case generation. They have also been adapted to other languages. . . ”*

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<sup>†</sup>See [www.sigplan.org/Awards/ICFP/](http://www.sigplan.org/Awards/ICFP/).

# Testing with QuickCheck

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An open source library

QuickCheck on Hackage.<sup>†</sup>

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<sup>†</sup><http://hackage.haskell.org/package/QuickCheck>.

# Testing with QuickCheck

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Commercialisation

QuviQ ([www.quviq.com/](http://www.quviq.com/)).

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## Adaptations

QuickCheck has been ported to various languages (Wikipedia 2025-07-28).

|             |             |            |         |           |
|-------------|-------------|------------|---------|-----------|
| C           | C#          | C++        | CHICKEN | CLOJURE   |
| COMMON LISP | CoQ         | D          | ELM     | ELIXIR    |
| ERLANG      | F#          | FACTOR     | Go      | Io        |
| JAVA        | JAVASCRIPT  | JULIA      | LOGTALK | LUA       |
| MATHEMATICA | OBJECTIVE-C | OCAML      | PERL    | PROLOG    |
| PHP         | PONY        | PYTHON     | R       | RACKET    |
| RUBY        | RUST        | SCALA      | SCHEME  | SMALLTALK |
| STANDARD ML | SWIFT       | TYPESCRIPT | VB.NET  | VHILEY    |

# Testing with QuickCheck

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## False positive

The program works properly but the test pointed out a fail:

- There is a bug elsewhere.
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## False negative

There is a bug in the program but the test passed.

Recall Dijkstra's 1969 famous quote:

*"Program testing can be used to show the presence of bugs, but never to show their absence!" (Dijkstra 1970)*



# Testing with QuickCheck

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## Example

See demo.

## References

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E. W. Dijkstra (1970). Structured Programming. In: Software Engineering Techniques (NATO Software Engineering Conference 1969). Ed. by J. N. Buxton and B. Randell, pp. 84–88 (cit. on pp. 7, 8).